



Analog Stimulus Isolator

2200

INSTRUCTION MANUAL

FOR

ANALOG STIMULUS ISOLATOR

MODEL 2200

Serial # _____

Date _____

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Contents

General Description	1
Operating Instructions	2
Examples and Applications	6
Theory of Operation	10
Specifications	11
Warranty and Service	14

Each Stimulus Isolator is delivered complete with:

Battery Charger

NOTE

This instrument is not intended for clinical measurements using human subjects. A-M Systems does not assume responsibility for injury or damage due to the misuse of this instrument.

General Description



The *Model 2200 Analog Stimulus Isolator* provides opto-electrical isolation for stimuli generated by a wide range of signal sources. The signal is DC-coupled and can take on any waveshape within its wide bandwidth. The input signal voltage, scaled according to the Range switch setting, can result in an output as large as 100V (or $\pm 50V$), and $\pm 5mA$. The output signal can be turned on and off by a digital Control Gate. If for any reason the instrument cannot follow the input signal, an Error indicator lights.

The isolated output section is battery-powered and optically-coupled to the input section, for the ultimate in clean isolation. The battery is adequate to provide full function for at least 8 hours of continuous use. An internal battery tester is provided. A charger is supplied, and an oven-night charge cycle is all that is required to ready the 2200 for another day's use.

The instrument is well suited to accepting analog signals generated under computer control, isolating these signals from the noisy electrical environment associated with a computer, and coupling them to biological tissue.

NOTE

The instrument is not intended for clinical measurements using human subjects. A-M Systems, Inc. does not assume responsibility for injury or damage due to the misuse of this instrument.

WARNING the Model 2200 can produce potentially dangerous voltages (up to 100V) at the output. The isolated output is not connected internally to safety (earth) ground in any way. Some part of the external circuit should be connected to safety ground. Use caution in handling any wires, connectors, or electrodes which may be directly or indirectly attached to the Model 2200 output. Some kinds of connectors (e.g. BNC connectors) have exposed metal parts which may float at dangerous potentials unless externally connected to safety ground. **For greatest safety, turn the Model 2200 off before handling connections to the output.**

Operating Instructions

The instrument is very simple to operate. First, check the battery condition: set the Power switch to off, and push the Battery Test button. The Test light should come on, indicating an adequate charge. Connect your input signal to the Signal In connector. Set the Polarity control to the predominant polarity of the waveform. Connect the Output to the stimulation electrode (or other load you wish to drive) and set the Range switch to the correct scale. If you wish to turn the output signal on or off by a separate controller (e.g. a timing signal), or be able to test the batteries via computer control during the course of an extended experiment, connect the control signal to the Control/ Gate input. Now turn the Power switch to On.

The output may go to an unpredictable state if the battery charge is not maintained. It is the user's responsibility to ensure an adequate battery charge. Regular use of either front-panel or control-port battery testing is sufficient to guarantee proper battery charge.

The Error indicator should remain unlit in normal operation. The Error LED will light when the computer-interface battery test is in operation. Be aware, however, that the Error LED will not light if the batteries are dead! It is normal for the LED to turn on briefly when the front panel controls are switched.

To recharge the batteries, connect the battery charger to the rear Charger input, and set the Power switch to Charge. Note that if the switch is left in the Charge position without any incoming electrical power, the batteries will very slowly, but certainly, become discharged.

Signal Control

The voltage at the Signal In BNC defines the output waveform. The signal is DC coupled across the optical isolation barrier, so virtually any waveform can be used. It can be turned on and off with a digital Gate signal through the Control port

Set the Range control to achieve the type (voltage or current) and amplitude of output required. The output power is limited to 100V (± 50 V in bipolar mode) and ± 5 mA. For applications requiring the highest voltages (for example, trying to push "large" currents through high resistance electrodes), it may be necessary to set the Polarity switch to the dominant polarity of your signal. For most users, it should be necessary to set the Polarity switch to the dominant polarity of your signal. For most users, it should be adequate to leave the switch set to Bipolar, giving equal range for both positive- and negative-going signals. For the most challenging cases, you may double the output power by combining two units.

NOTE: To avoid unintended transient output, refrain from changing the RANGE setting during an experiment. If it is necessary, the proper steps would be

- 1) *disconnect animal / prep*
- 2) *turn off POWER*
- 3) *change RANGE*
- 4) *turn on POWER*
- 5) *reconnect animal / prep*

For users requiring the lowest possible offset voltage or current, it may be necessary to readjust the Offset. Use the small insulated screwdriver (supplied with the Model 2200) to zero the output while the input voltage is set to zero. It is best to do this with the Range switch set to the desired position.

Battery Testing and Maintenance

There are two internal rechargeable batteries: a small 9V NiMH battery used to power the input section; and a larger gelled-electrolyte 12V battery used to power the output. The batteries are designed to last at least 8 operating hours from a full charge. Brand-new batteries often have a slightly reduced capacity, which improves after a few charge-discharge cycles.

To test the state of the batteries, simply turn the Power to Off, and push the front panel Battery/Test push-button. The Battery/OK LED will light if both battery voltages are sufficiently high. Warning: this light does not indicate how long the charge will remain high. It is up to the user to maintain the battery charge properly.

To preserve maximum battery life, use the full battery capacity. Do not routinely discharge the battery completely as the gelled-electrolyte (12V) battery life will eventually deteriorate if you discharge it too far. At the other extreme, do not frequently discharge the batteries to a small fraction of their capacity: shallow discharging of the unit may lead to diminished charge life of the input-side (9V) battery.

If battery charge life is a problem, the simplest solution is to keep on hand a supply of one or more sets of fully charged replacement batteries in addition to the internal set. - It is easy to change the batteries: turn the Power switch Off, turn the unit upside-down, and unscrew the two recessed Phillips head bolts (see figure 1). While holding the unit together, turn the unit right-side up. The top side of the case comes off with an easy vertical pull. The 9V battery is in a battery holder on the horizontally-mounted circuit board; simply pry the discharged unit out (pull on the attached tab), and replace with another battery. Disconnect the wire from the larger 12V battery at the connector to the circuit board. Now pull the battery from the Velcro® pad that it is sitting on. Replace the battery, making sure that it is firmly pressed into the pad. Reconnect the new battery to the circuit board, making certain that the keyed connector is locked together with the correct orientation! If the connector does not go together easily, you

may have it upside-down. Reinstall the top cover and tighten the two bolts. Pushing the Test push button should now light the Battery OK LED.

NOTE: If the Model 2200 is beeping, then the 9V battery needs to be replaced.

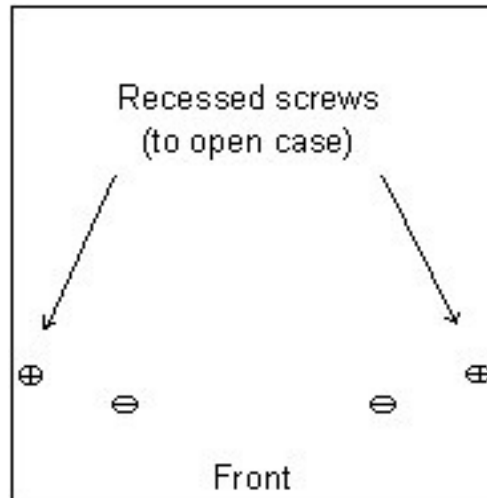


Figure 1. Bottom view of case: screw locations

If for some reason it is necessary to store the 2200 for more than a few months, it is recommended that the unit be kept at a low temperature (5-10°C). Before storing for longer than a year, it is advisable to remove the batteries from the unit; store the batteries separately

Control Port

Digital control is performed via the subminiature DB-9 connector on the instrument.

Signal Description

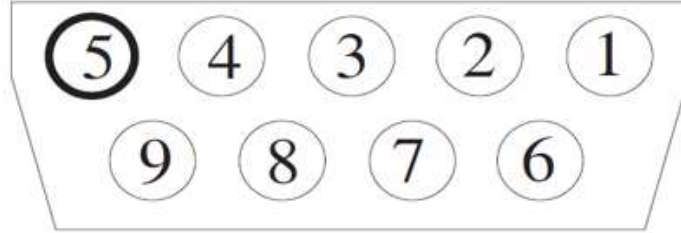
The output signal is turned on and off via the Gate line. The default (open circuit) condition is a logical "1", allowing unimpeded signal transmission from input to output.

Output errors may be detected via the Error line. Normally at a logical "0", this line goes high for at least 500ms if the output amplifier cannot drive the load satisfactorily or if the output range were exceeded, such as trying to obtain 200V from the unit.

Since it is possible that either or both of the internal batteries will be too low to provide a readable Error signal, it is strongly recommended that a periodic battery check be done via the front panel button in order to ensure proper operation.

Pinout

The pins are connected as follows. Note that while this is mechanically compatible with a 9-pin RS-232 connector, it is not electrically compatible. The 2200 is not smart enough to understand serial communications; and for some, the slow, erratic speed of serial communications would compromise performance



<i>Pin Number</i>	<i>In/Out</i>	<i>Pin Name</i>
1	O	Battery OK
2	O	Output Error
3		Ground
4	I	Gate
5	O	VCC (do not connect to any other device)
6	I	Battery Test
7		Ground
8		No Connection
9		No Connection

All signals are active-high.

The accessory cable #851000 gives a connection to the Gate signal (and ground) only

Do not connect the Battery/Test signal permanently low: this would reduce battery charge life, and prevent real output errors from being detected. Do not connect anything to pin 5 of DB-9 connector without consulting A-M Systems, or its representative.

Examples and Applications

The first thing that should be done in any use of this instrument is to check the batteries using the internal battery test function. Set the Power switch set to Off, and push the Battery/Test push-button. The Battery OK light should come on, indicating an adequate charge. If the test light does not come on, the batteries need to be recharged. The following applications assume that this simple test has been passed first.

Monophasic Output Pulse

An example of a simple application (and basic instrument test) is using a pulse generator, function generator, or other source of TTL-compatible voltages to provide an On/Off signal.

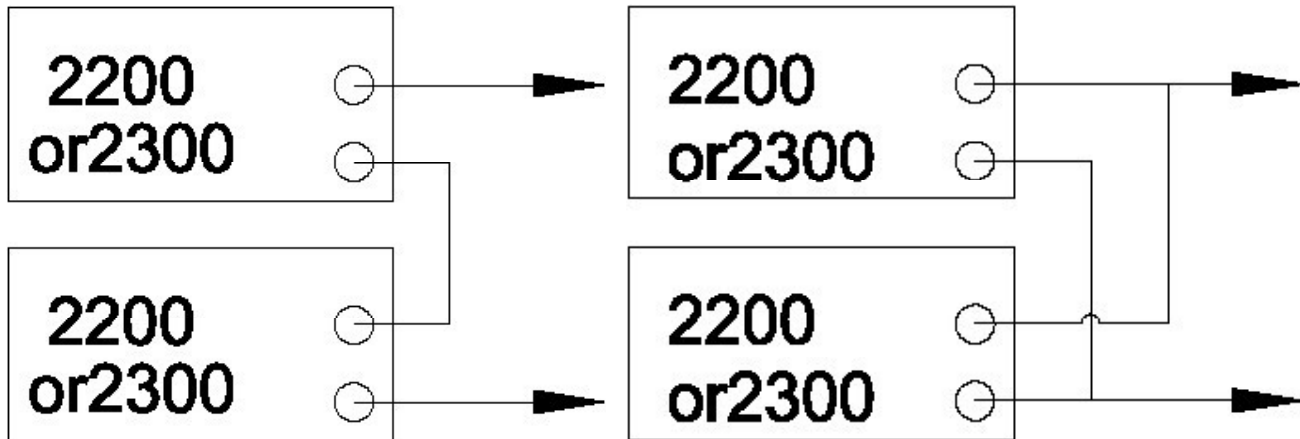
<i>Control/Connector</i>	<i>Setting</i>
Output	Connect a 10kohm resistor to the output Connect an oscilloscope probe across the resistor.
Input	Connect a function generator with a 1kHz 1V p-p sine wave
Sign	Bipolar
Range	1V/V
Power	On

You should observe the same waveform on both input and output. Rotating the DC offset control on the function generator should cause the DC level to change equally in input and output.

Change the Range control to 1mA/V. The waveform should stay the same. Now turn the Power off, disconnect the 1kW resistor, and turn the Power back on. The output waveform should be much larger (flattening against the power supply rails), and the error LED should light up as the 2200 fails to drive up to 6500mA into the high impedance of the oscilloscope input/load.

Combing Two 2200's For Increased Output

If you require up to twice the maximum output voltage, connect two 2200 units in a series. If you require up to twice the maximum current of a single 2200, connect the 2200's in parallel. The Gate inputs may be connected (and driven) together, by the same timing source.



Problem Solving

If the instrument appears to be not working properly, check all of the control settings and connections. The following brief summary of typical problems for the user (along with the most common solutions may help:

The Instrument is beeping

Replace the 9V battery and reset the alarm button on the rear panel

No output (and the Error LED does not light).

- One or both of the batteries are uncharged (test batteries: recharge as required).
- The Gate input is held low (must be high for an output to occur).

Battery OK LED does not light when pressing Test.

- Power switch is not set to Off (set to Off).
- One or both of the batteries need recharging.

Battery OK LED does not light, but Battery OK and Error signals of port test positive.

- This indicates that the batteries are almost fully discharged. The front panel battery test requires a slightly greater battery level to operate successfully.

Error LED on, even with no input

- Low batteries (recharge).

Error LED lights when Gate is turned on

- Excessive drive: Unit is limited to $\pm 100V$ in monopolar mode or $\pm 50V$ in bipolar mode (reduce input drive or gain).
- Current mode: load is disconnected (check wiring); or load impedance is too high for limited output voltage capability (cascade more than one unit).
- Battery charge is low (recharge)
- Gate signal erroneously connected to Battery Test input (change connection).

Error LED lights when Battery Test signal is used

- This is normal.

Excessive noise on the output

- Charger is still plugged in (unplug/disconnect).
- In the lowest current range: stray electrical interference can degrade noise performance if it mostly couples into the

output pin. (Try operating the 2200 within a Faraday (electrostatic shielding) cage. Match or reduce the stray capacitance to each output pin. Shorted the output connections. If these are inadequate, reverse the output banana connector, and invert the signal polarity).

If you suspect the Model 2300 may have failed, we recommend checking the basic level of functionality. If the instrument fails at this basic level, the instrument is defective and must be recalibrated or repaired.

Theory of Operation

The operation of the Model 2200 is summarized in the block diagram (see Fig. 4). The input signal is buffered, then enabled or disabled by an analog gate. When disabled, the output voltage at this stage is zero. A linear optoisolator system is used to couple the signal to the isolated section. The isolated signal is amplified (and possibly converted to a current) according to the Range setting.

The output amplifier is connected as a voltage-gain amplifier in voltage modes, with different feedback resistances depending on the setting. In the current mode, the same amplifier is used to fix a virtual ground within the instrument. The two currents connected to this virtual ground are the set current, determined by the range switch and input signal amplitude; and the load, driven by the amplifier output

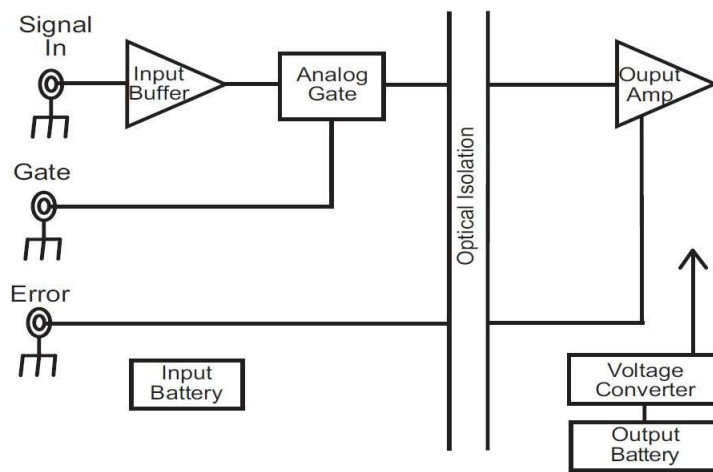


Figure 4. Instrument block diagram

With the Control Gate input signal open (disconnected), or set to a logical 1 by an external signal, the signal propagates normally as described above. If the Gate is set to a logical 0, the internal signal is connected to the input ground instead of the input signal.

Output errors are measured by sensing the output amplifier's differential input voltage. If for any reason the amplifier fails to deliver the requested signal, a warning LED illuminates and the Error output pin is set to a TTL-high level.

Two sets of batteries are used to power the 2200. Note that the BATTERY tester only detects the present state of the batteries, and does not indicate how long the charged batteries will be able to support the load. The batteries used in the 2200 vary more from unit to unit than from fully-charged to nearly-discharged, making it impractical to measure the charge state. It is recommended that you, the user, keep track of battery usage to ensure that the battery does not become exhausted at a critical time.

Specifications

There are three classes of specifications. Class A specifications are tested in all units, and are guaranteed. Class B specifications are inherent in the design; they are indirectly tested, and are guaranteed. Class C specifications are typical operating values which are occasionally tested; these are typical values, given for your information but not guaranteed. All specifications require that the batteries are adequately charged, and the instrument is fully warmed up (at least 15 minutes).

Input

Input resistance	B	1 Mohm $\pm$ 1% in parallel with less than 22pF
Logical '0'	B	<0.8 V
Logical '1'	B	>2.4 V
Input equivalent circuit	C	>33ohms in parallel with no more than 35pf
Maximum linear input	B	$\pm$ 10V
Signal Input to output delay	C	$\approx$ 5 μ s for a load of more than 100kohms in parallel with 50pF
Control timing jitter	C	<0.1 μ s
Gate polarity	A	active high (default is high = on)
Gate to output delay	C	$\approx$ 6 μ s
Battery Test polarity	A	active high (default is low = off)
Battery Test to output delay	C	<0.1 ms (turning on); < 1 sec (turning off)

Outputs

Maximum output capability	A	$\pm$ 50 V Bipolar (+ polarity) or $\pm$ 100V Monopolar. (- polarity) $\pm$ 5mA. Both limits apply to all Range settings.
Full scale output	B	$\pm$ 10V x Range setting
Gain	A	Within $\pm$ 3% of setting at DC. Accuracy is reduced for high

		output currents in voltage modes, to $\pm 2\%$ at full output	
Transient response	A	<10μs risetime into 1megohm (10 kohm in current modes) in parallel with less than 47pF. Decreases to $\approx 4 \mu$s with jumper J350 removed.	
Bandwidth	B	Within 1-3 dB from DC to 40kHz (same load impedances as above. Increases to ≈ 100kHz by removing jumper J350.	
Maximum Slew rate (the effective bandwidth is reduced for large signals) with 10kohm and 20pF load.	C	RANGE	SLEW RATE
		10V/V	5V/ μ s
		1V/V	0.6V/ μ s
		1mA/V	0.4mA/ μ s
		0.1mA/V	54 μ A/ μ s
		10 μ A/V	5.6 μ A / μ s
		10A /V	0.6 μ A / μ s

Note: Risetime in current modes is slower than in voltage modes if the wiring capacitance approaches or exceeds $4.5 \times 10^6 R^{-1} \text{pF}$ (R in ohms). Bandwidth is decreased by the same factor. This is a function of the load impedance driven by a constant current source, not a static property specific to the Model 2200.

Offset	A	within $\pm 0.002\%$ of full scale (plus additional $\pm 1\text{nA}$ offset in current modes) at 25°C	
Drift	B	Less than $\pm 0.01\%$ of full scale per $^{\circ}\text{C}$ from 10 to 30°C	
Output impedance	C	RANGE	SLEW RATE
		10V/V	≈ 50 ohms
		1V/V	≈ 50 ohms
		1mA/V	$>1\text{gigohm}$
		10 $\mu\text{A/V}$	$>50\text{gigohm}$
		1 $\mu\text{A/V}$	$>50\text{gigohm}$

Noise, typical (voltage modes: open circuit load; current modes: 100kohm load) 10 Hz - 100 kHz bandwidth. J350 in place	C	MODE	RMS % FULL SCALE	
		10V/V	<10mV	0.01
		1V/V	<3mV	0.03
		1mA/V	<1µA	0.01
		0.1mA/V	<100µA	0.01
		10µA/V	<10nA	0.01
		1µA/V	<2na	0.02

Isolation resistance A >>200Mohms

Isolation capacitance C <20pF

Overload error C Flat to approx. 10ms duration
Decreasing sensitivity for
shorter pulses.

Miscellaneous

Battery load life (from full
charge) B At least 8 hours (Control/Battery
Test on less than 100 seconds,
total)

Battery charge time B No more than 14 hours

Battery charge life (unit off) B At least 2 months

Operating temperature
ange 10-30°C

Operating humidity 5-80% RH

Storage temperature range 0-40°C

Dimensions 6.35 cmH x 15.4 cmW x 15.9 cmD

Weight 1.15kg

Warranty and Service

LIMITED WARRANTY

What does this warranty cover?

A-M Systems, LLC (hereinafter, “A-M Systems”) warrants to the Purchaser that the Instruments manufactured by A-M Systems (hereinafter the “hardware”), and sold after June 1, 2025, is free from defects in workmanship or material under normal use and service for 5 years from date of purchase. Headstages manufactured by A-M Systems and sold after June 1, 2025, will be repaired under warranty only once per year. This warranty commences on the date of delivery of the hardware to the Purchaser.

For hardware sold prior to June 1, 2025, the warranty in effect at time of purchase applies.

What are the obligations of A-M Systems under this warranty?

During the warranty period, A-M Systems agrees to repair or replace, at its sole option, without charge to the Purchaser, any defective component part of the hardware provided that the defective part can be purchased from mainstream, common, electronic component distributors such as Digi-Key Electronics, Newark, or Mouser Electronics. To obtain warranty service, the Purchaser must return the hardware to A-M Systems or an authorized A-M Systems distributor in an adequate shipping container. Any postage, shipping and insurance charges incurred in shipping the hardware to A-M Systems must be prepaid by the Purchaser, and all risk for the hardware shall remain with Purchaser until A-M Systems takes receipt of the hardware. Upon receipt, A-M Systems will promptly repair or replace the defective unit and then return the hardware (or its replacement) to the Purchaser with postage, shipping, and insurance prepaid by the Purchaser. A-M Systems may use reconditioned or like-new parts or units at its sole option, when repairing any hardware. Repaired products shall carry the same amount of outstanding warranty as from original purchase. Any claim under the warranty must include a dated proof of purchase of the hardware covered by this warranty. In any event, A-M Systems liability for defective hardware is limited to repairing or replacing the hardware.

What is not covered by this warranty?

This warranty is contingent upon proper use and maintenance of the hardware by the Purchaser and does not cover batteries. Neglect, misuse whether intentional or otherwise, tampering with or altering the hardware, damage caused by accident, damage caused by unusual physical, electrical, chemical, or electromechanical stress, damage caused by failure of electrical power, or damage caused during transportation are not covered by this warranty. Further, no guarantee is made regarding software compatibility with future updated operating systems. Products may not be returned to A-M Systems for service, whether under warranty or otherwise, which are contaminated by infectious agents, radioactive compounds or other materials constituting a health hazard to employees of A-M Systems

What are the limits of liability for A-M Systems under this warranty?

A-M Systems shall not be liable for loss of data, lost profits or savings, or any special, incidental, consequential, indirect or other similar damages, whether arising from breach of contract, negligence, or other legal action, even if the company or its agent has been advised of the possibility of such damages, or for any claim brought against you by another party.

THIS EQUIPMENT IS NOT INTENDED FOR CLINICAL MEASUREMENTS USING HUMAN SUBJECTS.

A-M SYSTEMS DOES NOT ASSUME RESPONSIBILITY FOR INJURY OR DAMAGE DUE TO MISUSE OF
THIS EQUIPMENT.

Jurisdictions vary with regard to the enforceability of provisions excluding or limiting liability for incidental or consequential damages. Check the provision of your local jurisdiction to find out whether the above exclusion applies to you.

This warranty allocates risks of product failure between the Purchaser and A-M Systems. A-M Systems hardware pricing reflects this allocation of risk and the limitations of liability contained in this warranty. The agents, employees, distributors, and dealers of A-M Systems are not authorized to make modifications to this warranty, or additional warranties binding on the company. Accordingly, additional statements such as dealer advertising or presentations, whether oral or written, do not constitute warranties by A-M Systems and should not be relied upon. This warranty gives you specific legal rights. You may also have other rights which vary from one jurisdiction to another.

THE WARRANTY AND REMEDY PROVIDED ABOVE IS IN LIEU OF ALL OTHER WARRANTIES AND REMEDIES, WHETHER EXPRESS OR IMPLIED. A-M SYSTEMS DISCLAIMS THE WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR USE, WITHOUT LIMITATION..

Revision History

Rev	Date	Description
8	1/18/19	DCR 202615 Review content, add content rev control.
9	3/19/20	DCR 203316. Update warranty.
10	5/4/22	DCR 203652. Update language about range switch.
11	4/29/25	DCR 204239. Update Warranty